Renewable Energy Production

Analysis & Forecasting

Solar · Wind · Hydro · Geothermal | 2014 – 2025

Overview

This write-up summarises ten years of renewable energy production data (2014–2023) across four sources; Solar, Wind, Hydro, and Geothermal and presents a forecast through 2025. The goal is to make the key findings easy to understand without requiring a technical background.

Energy output is measured in gigawatt-hours (GWh). One GWh is roughly enough to power 330 homes for a month.

Key findings:

- Total renewable output more than doubled over the decade.
- Solar and Wind are the fastest-growing sources, each roughly tripling since 2014.
- Hydro is the largest single contributor but grows slowly.
- Geothermal is the most stable, consistent output all year round.
- By 2025, total output is forecast to reach approximately 19,860 GWh, up 13% from 2023.

1. Energy Production Over Time

The chart below shows monthly production for each source from 2014 to 2023. The dashed line is a 12-month rolling average that smooths out short-term fluctuations to reveal the underlying trend.

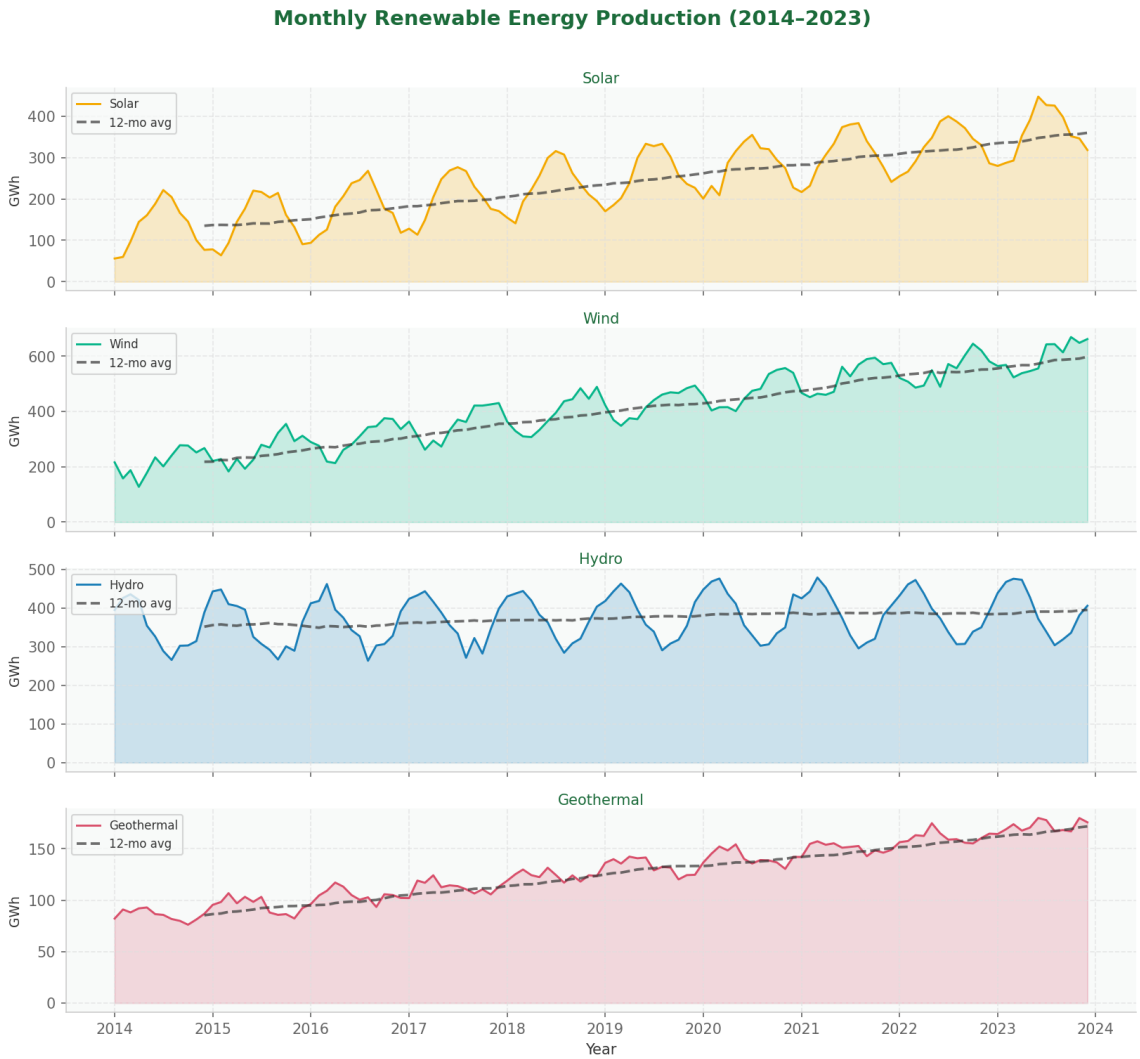

Figure 1 — Monthly energy output (GWh) per source, with 12-month rolling average (dashed).

All four sources show growth over the decade. Solar and Wind grow steeply, while Hydro and Geothermal grow more modestly. The seasonal ups and downs are very visible in Solar and Wind — more on that in the next section.

2. Annual Growth

Looking at total annual production makes the long-term growth story clearer. The stacked bars below show how the combined output of all four sources has increased year on year, and how the contribution of each source has shifted.

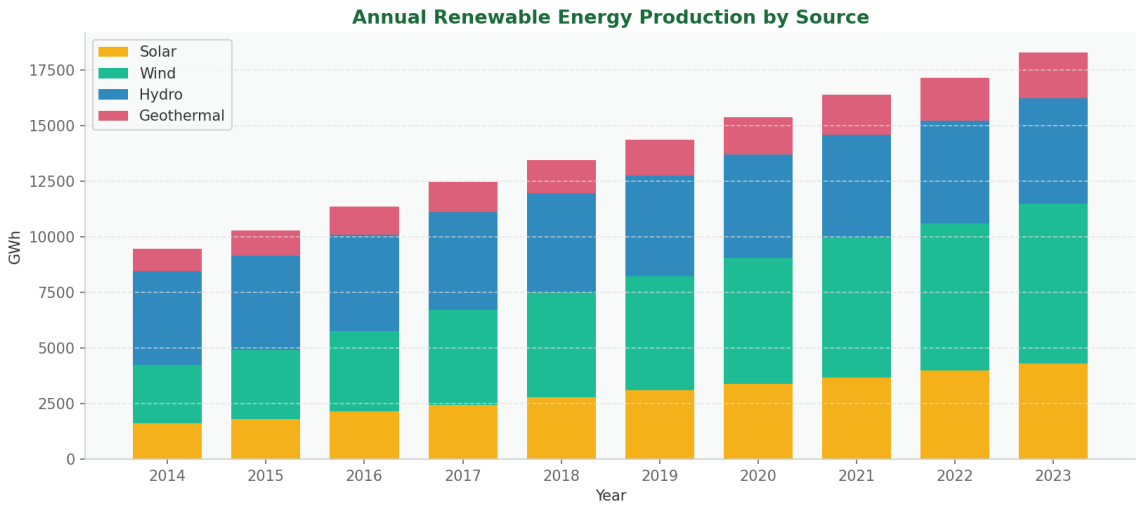

Figure 2 — Annual production by source (stacked), 2014–2023.

Wind and Solar (green and yellow) take up a growing share of the stack each year, reflecting significant investment in new capacity. Hydro (blue) remains the largest single block, though its share of the total gradually shrinks as the others grow faster.

3. Seasonal Patterns

Each energy source has its own seasonal rhythm, driven by the natural factors that power it. The charts below show the average output for each month of the year, averaged across all ten years of data.

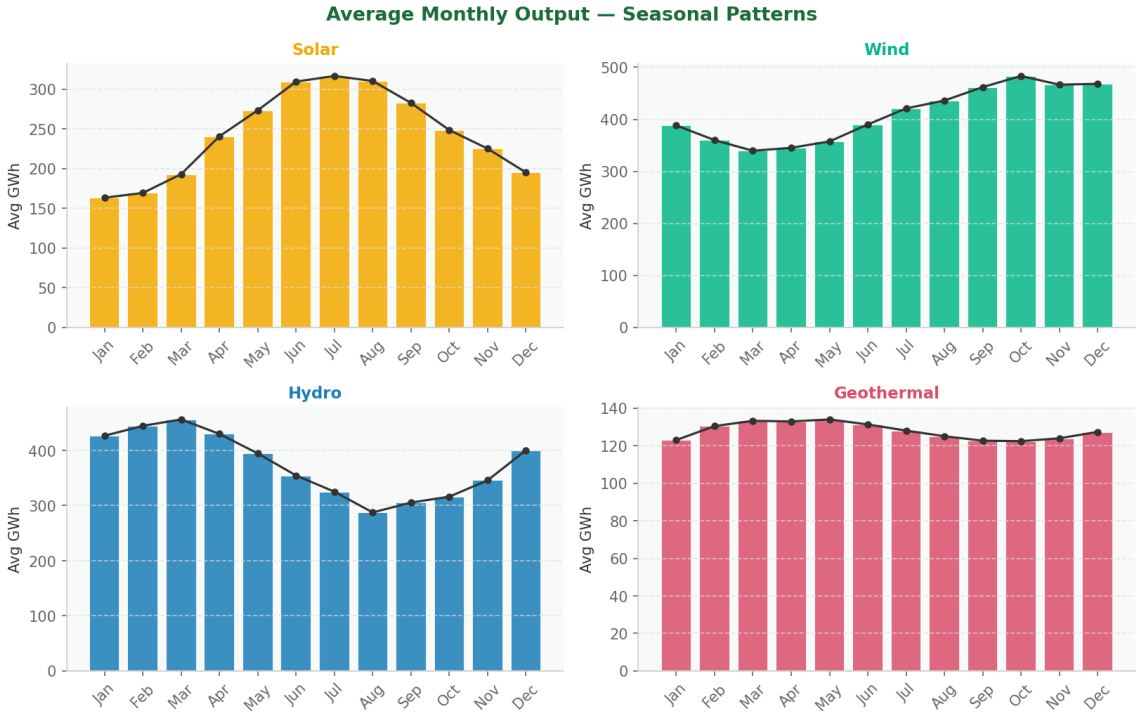

Figure 3 — Average monthly output by source, showing seasonal cycles.

What this tells us:

- Solar peaks sharply in summer (June–August) when days are longest, and drops in winter.
- Wind peaks in winter (November–January) when weather systems are strongest; a natural complement to Solar.
- Hydro peaks in spring (March–May) driven by snowmelt increasing river flows.
- Geothermal is almost flat across all months; its output does not depend on weather or seasons.

The fact that Solar and Wind peak at opposite times of year is important for grid planning: they naturally balance each other out across the calendar.

4. How the Energy Mix Has Shifted

The pie charts below compare the share of total annual output from each source in 2014 versus 2023. This shows how the composition of the renewable portfolio has changed over the decade.

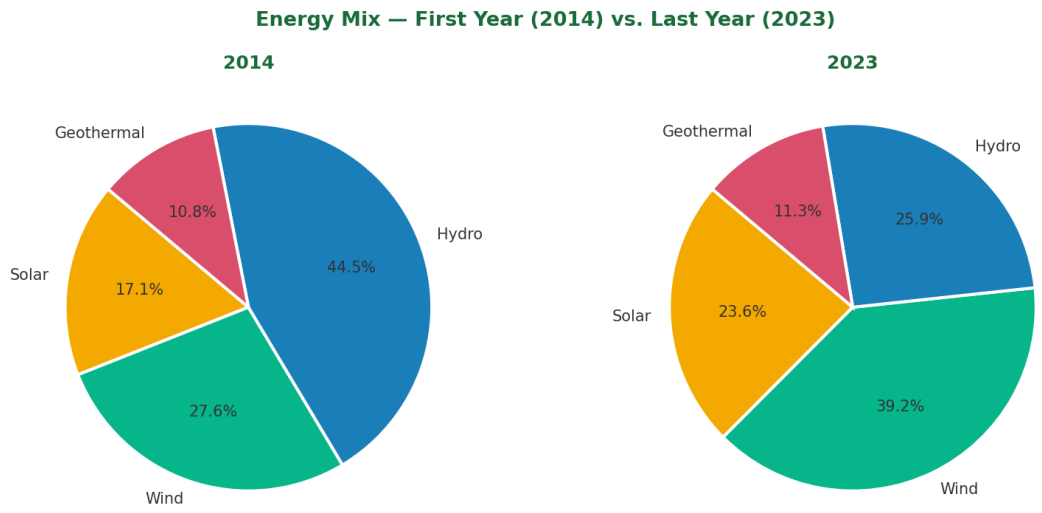

Figure 4 — Energy mix in 2014 (left) vs. 2023 (right).

Solar and Wind have both grown their share significantly. In 2014, Hydro dominated the portfolio. By 2023, the mix is more balanced, with Solar and Wind together accounting for a much larger proportion. This diversification reduces reliance on any single source and improves resilience.

5. Forecast: 2024–2025

Using a technique called Holt-Winters Exponential Smoothing — which captures both the long-term trend and the seasonal cycles of each source — we project output through the end of 2025. The shaded band around each forecast line represents the uncertainty range.

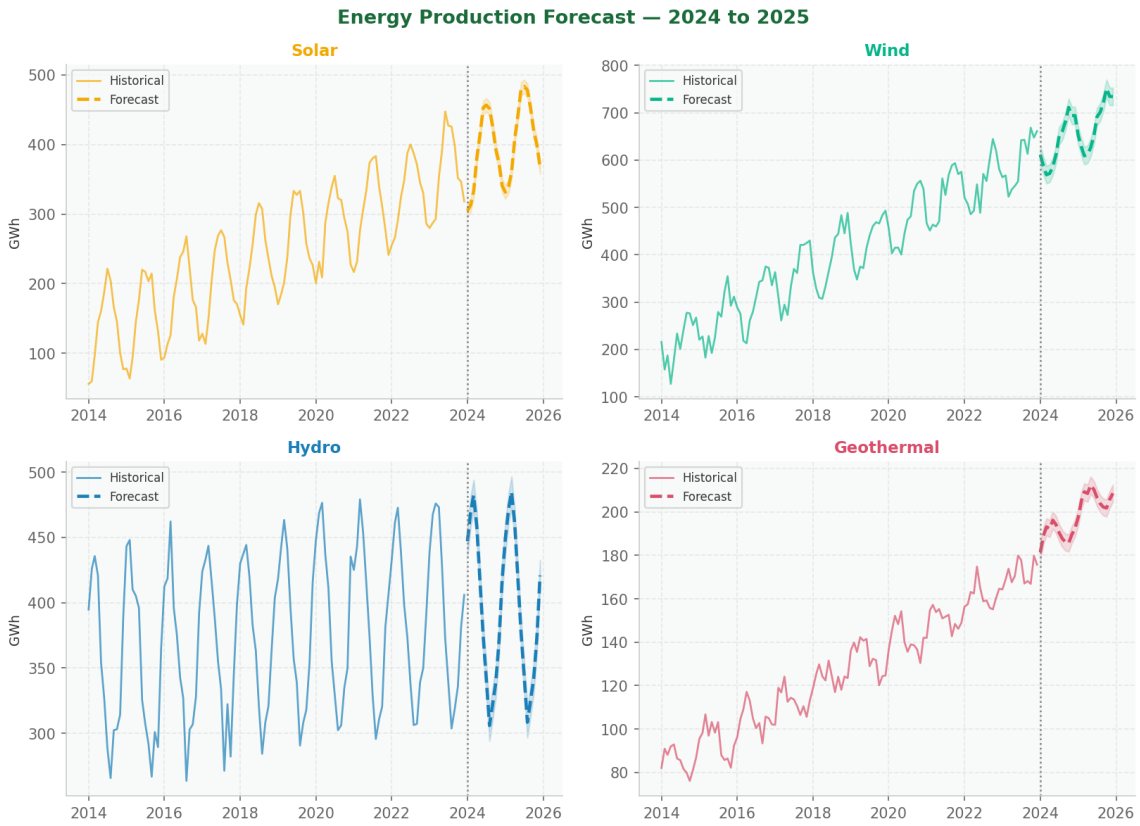

Figure 5 — Forecast output per source, 2024–2025. Shaded area shows uncertainty range.

Forecast summary for 2025 (approximate annual totals):

- Solar: ~5,740 GWh (+18% vs 2023)
- Wind: ~7,410 GWh (+18% vs 2023)
- Hydro: ~5,330 GWh (+3% vs 2023)
- Geothermal: ~1,380 GWh (+11% vs 2023)
- Total: ~19,860 GWh (+13% vs 2023)

Note: Forecasts are based on historical trend and seasonality. They do not account for unexpected events such as severe drought, major policy changes, or rapid shifts in installed capacity.

For questions or to request a deeper technical analysis, please contact me Email: olokotaiwoevthag@gmail.com.